

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

اللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَافِعًا،

وَرِزْقًا طَيِّبًا، وَعَمَلًا مُتَقَبَّلًا،

Expectation and Variance

Week 8

Expectation of variable

Let X be a random variable with probability distribution $f(x)$. The **mean**, or **expected value**, of X is

$$\mu = E(X) = \sum_x x f(x)$$

if X is discrete, and

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) \, dx$$

if X is continuous.

Example 2:

Let X be the no of heads in three tosses of a fair coin. Find $E(x)$.

Sample space $S=\{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

x	0	1	2	3
$P(x)$	$1/8$	$3/8$	$3/8$	$1/8$

$$E(X)=\sum xp(x)$$
$$=1.5$$

$$E(X) = \sum_{i=1}^n xp(x)$$

TASK

Find the probability distribution and expected value for X , if PMF for X can be defined as

$$f(x) = \frac{6 - |7 - x|}{36}, \text{ for } x = 2, 3, 4, \dots, 12$$

Task

Let X be the random variable that denotes the life in hours of a certain electronic device. The probability density function is

$$f(x) = \begin{cases} \frac{20,000}{x^3}, & x > 100, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected life of this type of device.

$$\mu = E(X) = \int_{100}^{\infty} x \frac{20,000}{x^3} dx = \int_{100}^{\infty} \frac{20,000}{x^2} dx = 200.$$

Expectation of function

Let X be a random variable with probability distribution $f(x)$. The expected value of the random variable $g(X)$ is

$$\mu_{g(X)} = E[g(X)] = \sum_x g(x) f(x)$$

if X is discrete, and

$$\mu_{g(X)} = E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) \, dx$$

if X is continuous.

Example: 3

Suppose that the number of cars X that pass through a car wash between 4:00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

x	4	5	6	7	8	9
$P(X = x)$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

Let $g(X) = 2X - 1$ represent the amount of money, in dollars, paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period.

$$\begin{aligned} E[g(X)] &= E(2X - 1) = \sum_{x=4}^9 (2x - 1)f(x) \\ &= (7) \left(\frac{1}{12}\right) + (9) \left(\frac{1}{12}\right) + (11) \left(\frac{1}{4}\right) + (13) \left(\frac{1}{4}\right) \\ &\quad + (15) \left(\frac{1}{6}\right) + (17) \left(\frac{1}{6}\right) = \$12.67. \end{aligned}$$

Task

Let X be a random variable with density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of $g(X) = 4X + 3$.

$$E(4X + 3) = \int_{-1}^2 \frac{(4x + 3)x^2}{3} dx = \frac{1}{3} \int_{-1}^2 (4x^3 + 3x^2) dx = 8.$$

Variance of random variable

Let X be a random variable with probability distribution $f(x)$ and mean μ . The variance of X is

$$\sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x), \quad \text{if } X \text{ is discrete, and}$$

$$\sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx, \quad \text{if } X \text{ is continuous.}$$

The positive square root of the variance, σ , is called the **standard deviation** of X .

Variance of X

$$V(X) = E(X^2) - \{E(X)\}^2$$

$$= \sum_{i=1}^n x^2 p(x) - \left\{ \sum_{i=1}^n x p(x) \right\}^2$$

Properties of $E(X)$

$$E(a) = a$$

$$E(aX + b) = aE(X) + b$$

$$E(X - \mu) = 0$$

$$E(X + Y) = E(X) + E(Y)$$

$$E(X - Y) = E(X) - E(Y)$$

$$E(XY) = E(X)E(Y)$$

$$E\left(\frac{X}{Y}\right) \neq \frac{E(X)}{E(Y)}$$

Properties of Variance

$$\text{Var}(X) \geq 0$$

$$\text{Var}(a) = 0$$

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$$

Task

The weekly demand for a drinking-water product, in thousands of liters, from a local chain of efficiency stores is a continuous random variable X having the probability density

$$f(x) = \begin{cases} 2(x-1), & 1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the mean and variance of X .

$$\mu = E(X) = 2 \int_1^2 x(x-1) \, dx = \frac{5}{3}$$

$$E(X^2) = 2 \int_1^2 x^2(x-1) \, dx = \frac{17}{6}$$

$$\sigma^2 = \frac{17}{6} - \left(\frac{5}{3}\right)^2 = \frac{1}{18}$$

TASK

Q: Let X have the following probability distribution:

x	1	2	3	4	5
$P(x)$	0.2	0.3	0.2	0.2	0.1

- Find the probability functions of X .
- Find the expectations of X .
- Find the variance of X .

TASK

Q: Let X have the following probability distribution:

x	1	2	3	4	5
$P(x)$	0.2	0.3	0.2	0.2	0.1

- Find the probability functions of $3X-1$, X^2 , $X^2 + 2$.
- Find the expectations of $3X-1$, X^2 , $X^2 + 2$.
- Find the variance of $3X-1$, X , $X + 2$.

TASK

Calculate the variance of $g(X) = 2X + 3$, where X is a random variable with probability distribution

x	0	1	2	3
$f(x)$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{2}$	$\frac{1}{8}$

Task:

Let X be a random variable with density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the variance of the random variable $g(X) = 4X + 3$.

Covariance

$$\text{Cov}(X, Y) = E[XY] - (EX)(EY).$$

Let X and Y be random variables with joint probability distribution $f(x, y)$. The covariance of X and Y is

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \sum_x \sum_y (x - \mu_X)(y - \mu_Y) f(x, y)$$

if X and Y are discrete, and

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x, y) \, dx \, dy$$

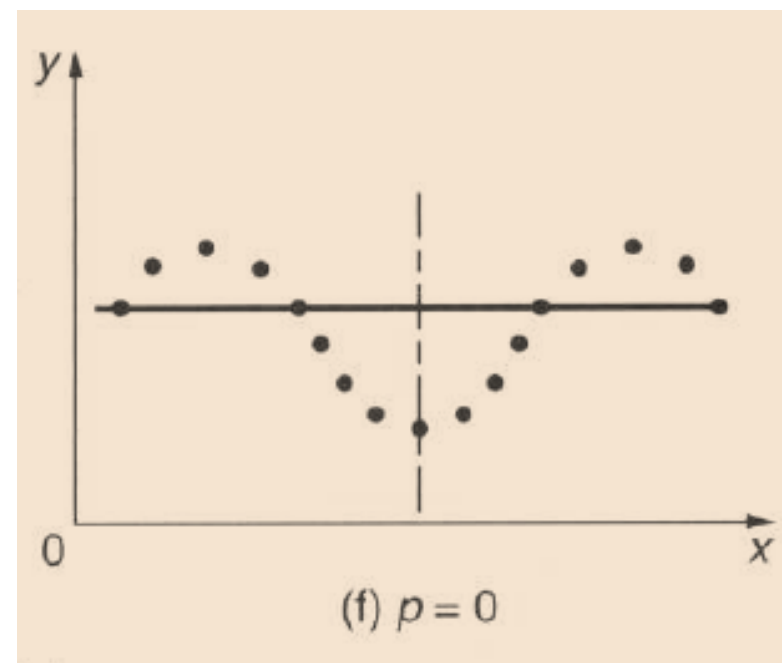
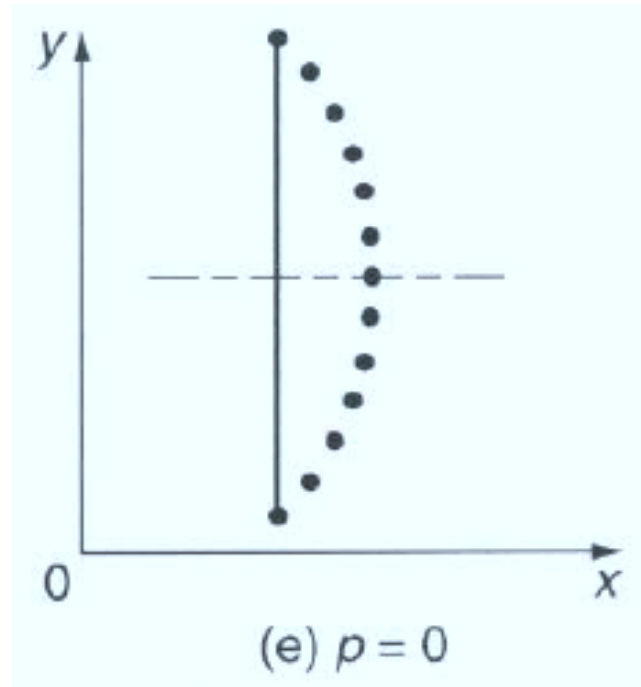
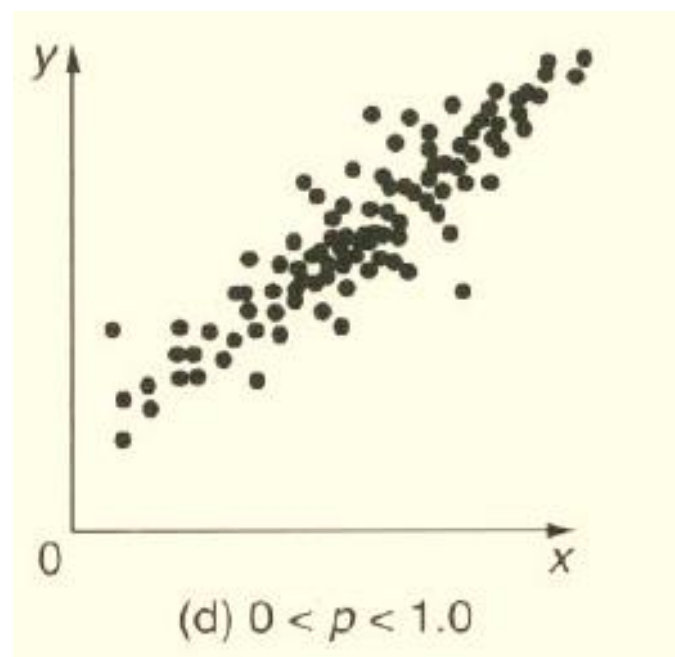
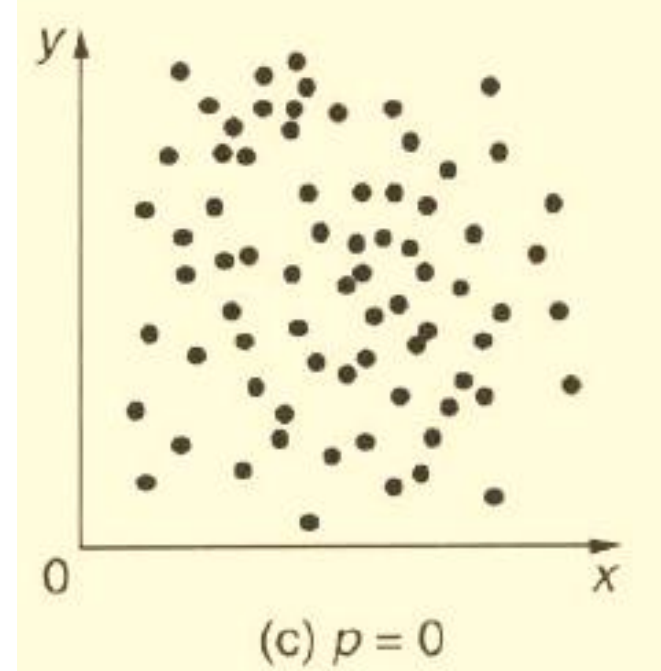
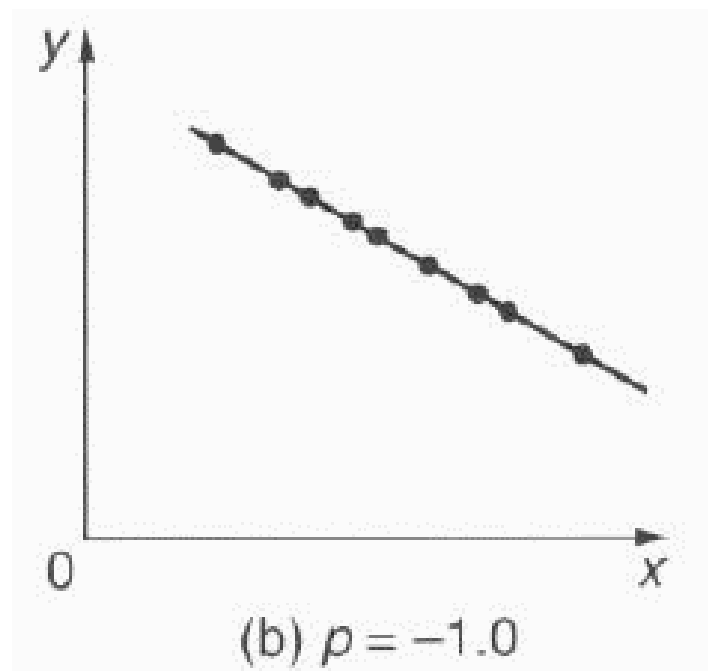
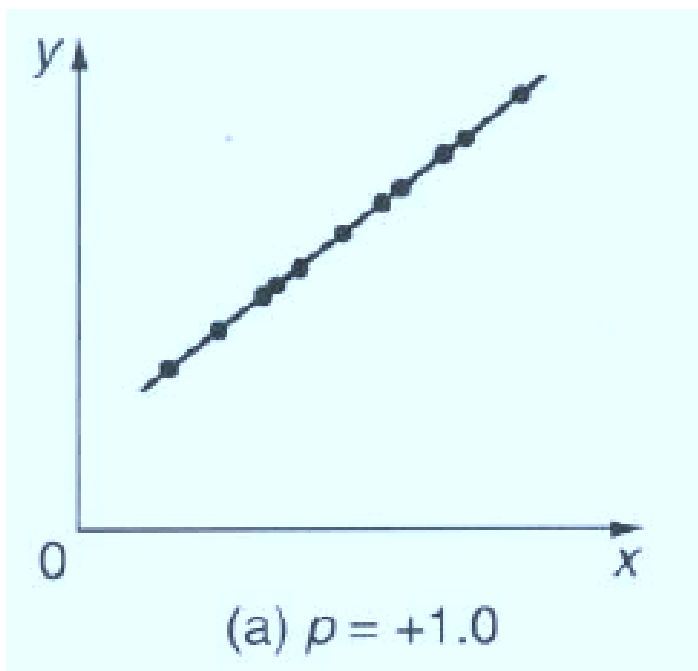
if X and Y are continuous.

Correlation coefficient

Let X and Y be random variables with covariance σ_{XY} and standard deviations σ_X and σ_Y , respectively. The correlation coefficient of X and Y is

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

$$-1 \leq \rho_{XY} \leq 1$$



Example: 4

The fraction X of male runners and the fraction Y of female runners who compete in marathon races are described by the joint density function

$$f(x, y) = \begin{cases} 8xy, & 0 \leq y \leq x \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the correlation coefficient between X and Y

$$E(X^2) = \int_0^1 4x^5 \, dx = \frac{2}{3} \text{ and } E(Y^2) = \int_0^1 4y^3(1 - y^2) \, dy = 1 - \frac{2}{3} = \frac{1}{3},$$

$$\sigma_X^2 = \frac{2}{3} - \left(\frac{4}{5}\right)^2 = \frac{2}{75} \text{ and } \sigma_Y^2 = \frac{1}{3} - \left(\frac{8}{15}\right)^2 = \frac{11}{225}$$

$$\rho_{XY} = \frac{4/225}{\sqrt{(2/75)(11/225)}} = \frac{4}{\sqrt{66}}.$$

شاہیں

کیا میں نے اس خاکِ دہاں سے کنارہ
دیا پاں کی غلوتِ خوش آتی ہے مجھ کو
نہ بار بہاری، نہ گلہیں، نہ بلبل
عمیادیاں سے ہے پرہیز لازم
ہوا کے دیا پاں سے ہوتی ہے کاری
حمام و کبوتر کا بھوکا نہیں میں
جھپٹنا، پلٹنا، پلٹ کر جھپٹنا
یہ پارس، یہ مہم پکوروں کی دنیا
مرا نیلگوں آسماں بیکراں

پرندوں کی دنیا کا درویش ہوں میں

کہ شاہیں بناتا نہیں آشیانہ